

University of Information Technology & Sciences (UITS)

Faculty of Science and Engineering

Department of Computer Science and Engineering

Program: B.Sc. in CSE

Mid-Term Examination, Spring 2025

Course Title: Digital Logic Design

Course Code: CSE0611215/CSE215/CSE201

Marks: 20

Time: 1(one) hour

(Answer all questions.)

- | Q. No. | Marks |
|--|-------|
| 1. a) i. Convert the hexadecimal number 2C6B.F2 to binary, octal, and decimal numbers. | [04] |
| ii. Given the two binary numbers $X = 1010101$ and $Y = 1000010$, perform the subtraction
(a) $X - Y$, (b) $Y - X$, by using 2's complement. | |
| b) Prove that
i. $x + xy = x$
ii. De Morgan's law (any one) by using 3 variables. | [03] |
| c) Reduce the given function to the minimum number of literals
$F(A, B, C, D) = B'D + A'BC' + ACD + A'BC$ | [03] |
| 2. a) Interpret the output of the given logic circuit and derive the truth table. | [02] |
| b) Express the following function in sum of minterm and product of maxterm
$F(W, X, Y, Z) = X'Z + W'Z + XZ$ | [05] |
| c) Simplify the following Boolean function by using the KARNAUGH Map
$F(A, B, C) = A'C + A'B + AB'C + BC$ | [03] |